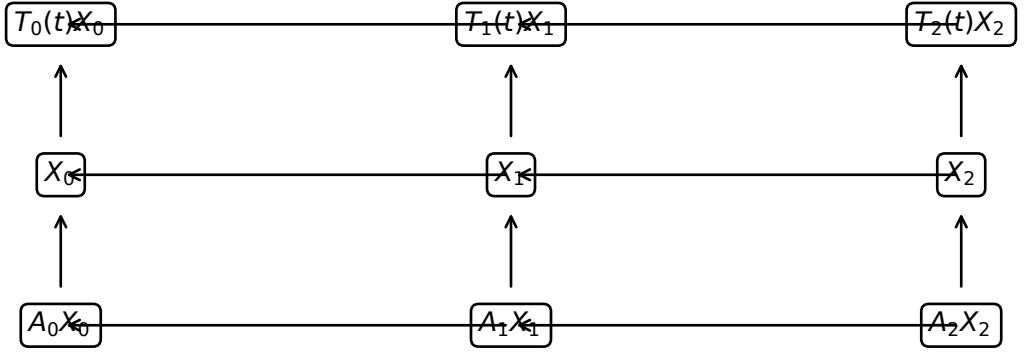

coarsening horizontally and evolution vertically